

IXOPO, South Africa

WMO: 685750

Lat: 30.1525S

Lon: 30.0747E

Elev: 937

StdP: 90.56

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.2	3.3	-6.3	2.5	14.4	-3.8	3.1	14.6	10.6	13.3	8.4	14.2	1.5	0	0.332

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.8	32.0	19.8	30.2	19.8	28.5	19.5	22.9	28.3	22.1	27.1	21.4	25.8	2.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	17.8	25.5	20.6	17.1	24.6	20.0	16.4	23.7	73.1	28.5	69.8	27.3	67.0	25.9	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.9	5.0	4.2	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.0	21.0	21.2	20.3	17.9	15.1	12.8	12.1	14.0	15.9	17.0	17.9	19.4
	DBStd	4.24	2.81	2.69	2.83	2.99	2.70	2.26	2.68	2.97	3.62	3.59	3.41	3.30
	HDD10.0	20	0	0	0	0	1	4	10	3	2	0	0	0
	HDD18.3	911	8	6	15	43	104	167	192	139	89	69	49	28
	CDD10.0	2580	340	315	318	237	159	87	77	126	178	217	237	291
	CDD18.3	429	90	87	74	30	3	0	1	4	16	27	36	60
	CDH23.3	2630	481	403	378	165	59	5	9	74	177	249	267	363
CDH26.7	783	159	121	107	35	8	0	0	19	52	83	88	111	
Wind	WSAvg	1.4	1.3	1.4	1.2	1.3	1.3	1.5	1.8	1.4	1.5	1.5	1.5	1.6
Precipitation	PrecAvg	781	122	90	93	49	17	20	25	27	44	89	89	122
	PrecMax	1039	220	187	172	119	71	118	191	92	175	166	190	304
	PrecMin	489	35	42	41	6	0	0	1	0	0	14	29	45
	PrecStd	154	42	34	40	28	18	28	41	26	35	41	36	59
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.3	33.0	32.2	30.0	28.0	23.7	24.5	29.1	32.1	32.7	32.9	33.1
		MCWB	21.7	22.0	19.9	18.1	14.4	12.0	11.6	13.8	15.6	17.0	18.6	21.9
	2%	DB	30.6	30.1	29.6	27.2	25.1	21.6	21.7	25.4	28.2	29.4	29.4	29.6
		MCWB	21.9	21.9	20.1	17.8	14.4	11.3	10.7	12.9	15.6	17.2	19.5	21.8
	5%	DB	28.2	27.7	27.2	25.0	22.8	19.7	19.7	22.5	25.3	26.4	26.6	27.2
		MCWB	21.4	21.5	20.2	17.6	14.4	11.0	10.3	13.1	15.5	17.5	19.1	21.0
	10%	DB	25.9	25.5	25.2	22.8	20.6	17.8	17.7	20.0	22.6	23.6	24.0	25.0
		MCWB	20.9	20.7	19.7	17.1	14.0	10.4	10.0	12.5	15.0	17.0	18.5	20.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.0	24.0	22.6	21.0	17.7	14.9	13.8	16.9	18.8	20.2	22.4	23.8
		MCDB	29.0	30.1	28.0	26.3	21.8	18.8	18.6	21.7	26.9	26.5	28.4	29.6
	2%	WB	22.7	22.6	21.5	19.3	16.4	13.4	13.0	15.6	17.4	19.2	20.8	22.5
		MCDB	27.9	27.8	26.3	24.0	21.2	17.3	17.2	21.1	23.7	25.4	25.8	27.9
	5%	WB	21.9	21.7	20.7	18.5	15.5	12.3	12.0	14.3	16.5	18.2	19.7	21.4
		MCDB	26.8	26.1	25.0	22.6	20.2	16.7	16.2	19.8	22.0	23.8	24.1	26.1
	10%	WB	21.1	21.0	20.0	17.7	14.6	11.4	11.2	13.4	15.6	17.3	18.8	20.4
		MCDB	25.2	24.6	23.9	21.4	19.0	16.1	15.5	18.2	20.4	22.1	23.0	24.4
Mean Daily Temperature Range	5% DB	MDBR	8.8	8.8	9.5	10.4	12.7	12.7	12.9	12.7	11.8	10.0	8.7	8.7
		MCDBR	13.5	12.5	13.4	14.3	16.2	15.4	16.2	17.6	18.2	16.6	15.3	13.7
		MCWBR	5.6	5.3	5.5	5.4	6.9	6.8	6.9	7.2	6.9	6.6	6.6	6.7
	5% WB	MCDBR	11.9	11.3	11.4	11.7	13.4	13.6	13.9	14.2	13.9	13.3	11.9	12.6
		MCWBR	5.8	5.4	5.7	5.4	6.5	7.2	7.1	6.5	6.4	6.4	6.2	6.7
Clear-Sky Solar Irradiance	taub	0.399	0.396	0.382	0.348	0.322	0.305	0.314	0.371	0.416	0.425	0.405	0.400	
	taud	2.356	2.363	2.383	2.440	2.473	2.496	2.461	2.308	2.198	2.216	2.303	2.343	
	Ebn at Noon	943	926	905	887	863	857	862	847	856	888	930	945	
	Edh at Noon	133	128	118	100	87	80	87	113	139	146	139	135	
All-Sky Solar Radiation	RadAvg	5.90	5.70	5.03	4.20	3.59	3.21	3.46	4.03	4.69	5.08	5.57	5.81	
	RadStd	0.46	0.45	0.27	0.26	0.19	0.17	0.20	0.24	0.29	0.45	0.42	0.38	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page